

Online Appendix:
GENERATIVE AI AT WORK

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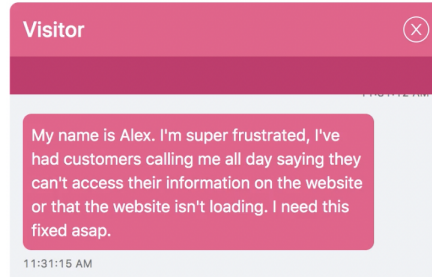
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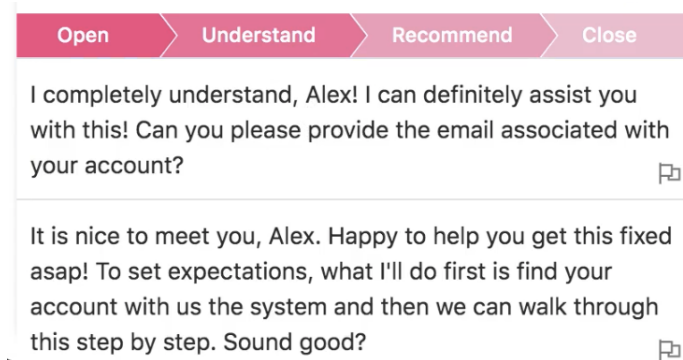
A. AI SYSTEM AND ROLLOUT

FIGURE A.I
Sample AI Output

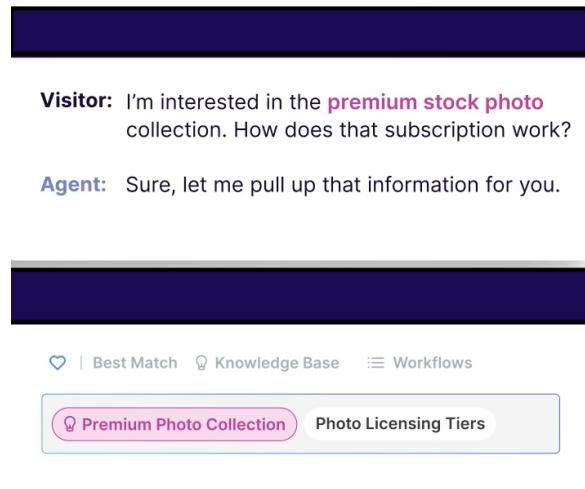
A. Sample Customer Issue



B. Sample AI-Generated Suggested Response

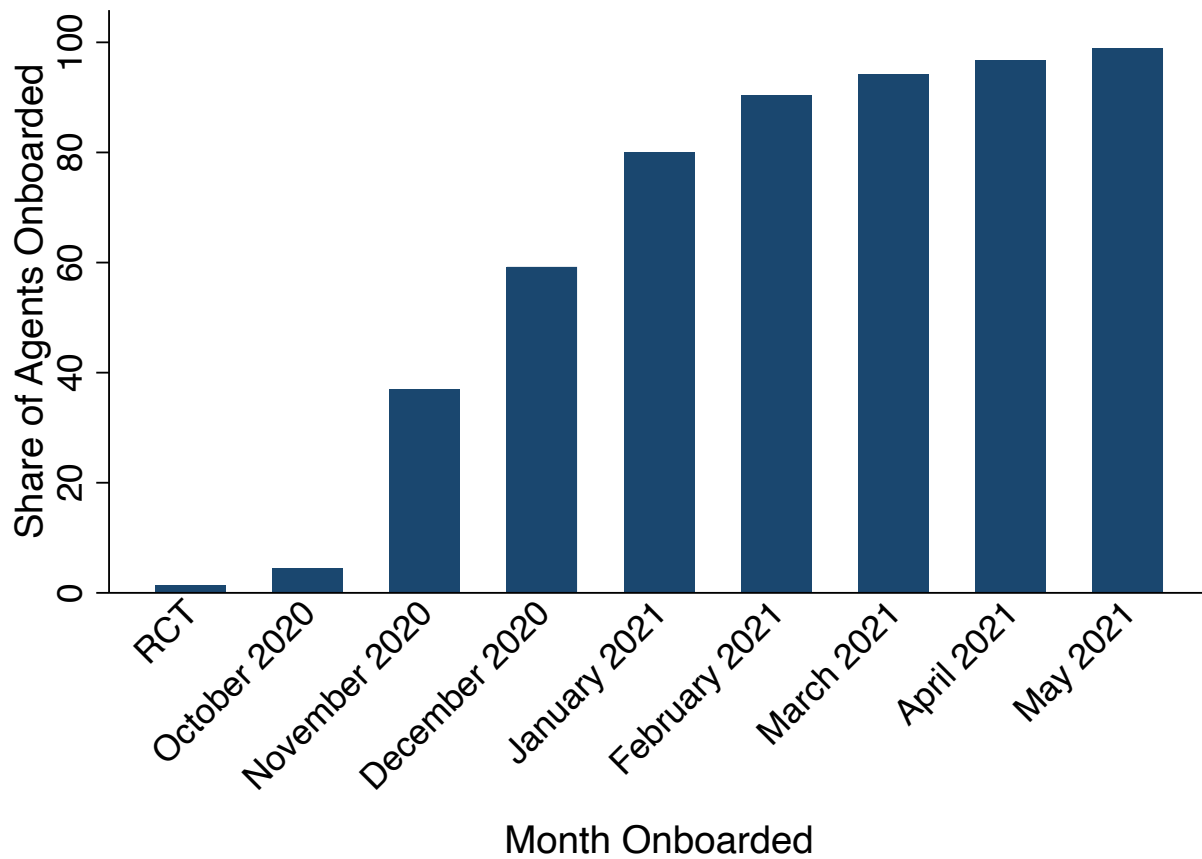


C. Sample AI-Generated Technical Link



NOTES: This figure illustrates AI-generated suggestions for customer service agents. Panel A shows a sample customer issue. Panel B displays AI-suggested responses for greeting and setting call expectations. Panel C presents an AI-recommended technical documentation excerpt from the company's internal knowledge base. Suggestions are only visible to agents, not customers, and agents can choose to use, modify, or ignore these suggestions when responding.

FIGURE A.II
Deployment Timeline



NOTES: This figure shows the share of agents deployed onto the AI system over the study period. Agents are deployed onto the AI system after a training session as described in Section IV.A. The small randomized control trial in August 2020 is analyzed in Section V.A.1. All data are from the firm's internal software systems.

B. AVERAGE PRODUCTIVITY EFFECTS

TABLE A.I
Randomized Control Trial Analysis

VARIABLES	(1) Res./Hr	(2) AHT	(3) Chats/Hr	(4) Res. Rate	(5) NPS
Post AI X Ever Treated	0.202** (0.0850)	-3.713*** (1.045)	0.105 (0.0718)	0.0169 (0.0246)	-1.393 (1.529)
Observations	6,998	15,100	15,100	6,998	7,176
R-squared	0.568	0.574	0.566	0.383	0.517
Year Month FE	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes
DV Mean	1.956	43	2.378	0.808	79.68

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

NOTES: This table focuses on the 22 agents who were on-boarded as part of the pilot randomized control trial (RCT). Because we do not have information on the specific agents (there were around 25) selected to be part of the control group, we compare the 22 pilot-treated agents with observations for all pre-treatment agents, controlling for agent, chat year-month, and months of agent tenure fixed effects. Data are at the agent-month level, and the standard errors are clustered at the agent level.

TABLE A.II
Main Effects: Company Team IV

VARIABLES	(1) Individual Treatment	(2) Res./Hr	(3) AHT	(4) Chats/Hr	(5) Res. Rate	(6) NPS
Earliest Team Treatment	0.311*** (0.0163)					
Post AI X Individual Treatment		0.550*** (0.0943)	-3.622** (1.444)	0.412*** (0.0791)	0.0740*** (0.0184)	-0.518 (0.946)
Observations	21,839	12,295	21,839	21,839	12,295	12,541
R-squared	0.813	0.177	0.102	0.115	0.007	0.007
Year Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	365.7					
Number of agent_id		2,163	3,633	3,633	2,163	2,214

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

NOTES: In this table we instrument agent level date of AI adoption with the minimum date of AI deployment on that agent's team. Column 1 shows the first stage regression of earliest team adoption on individual adoption, and Columns 2 through 6 show the 2SLS estimates on measures of productivity and call quality and efficiency. Regressions follow Equation 1 and include agent level, chat-year, and months of agent tenure fixed effects. Data are available at the agent-month level, and robust standard errors are clustered at the agent level. Information on the team is missing for some workers. Workers without this information are grouped into a single "missing" team.

TABLE A.III
Main Effects: Robustness to Alternative Clustering

VARIABLES	(1) Res./Hr	(2) Res./Hr	(3) Res./Hr
Post AI X Ever Treated	0.301*** (0.0329)	0.301*** (0.0455)	0.301*** (0.0498)
Observations	12,295	12,295	12,295
R-squared	0.575	0.575	0.575
Year Month FE	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes
DV Mean	2.176	2.176	2.176

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

NOTES: This table shows robustness of our Column 3 in Table II to different levels of clustering. Column 1 clusters our robust standard errors at the agent level; Column 2 clusters at the company/team level; and Column 3 clusters at the location (usually the agent's city of work). City of work may include call center locations for multiple companies (e.g., our data firm and a contracted outsourcing firm). Regressions follow Equation 1 and include agent level, chat-year, and months of agent tenure fixed effects. Data are available at the agent-month level, and robust standard errors are clustered at the agent level.

TABLE A.IV
Main Effects: Chat-Weighted

VARIABLES	(1) Res./Hour	(2) AHT	(3) Chats/Hr	(4) Res. Rate	(5) NPS
Post AI X Ever Treated	0.252*** (0.0318)	-3.100*** (0.315)	0.295*** (0.0263)	0.00325 (0.00725)	-0.119 (0.524)
Observations	12,295	21,839	21,839	12,295	12,541
R-squared	0.650	0.756	0.721	0.465	0.526
Year Month FE	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes
DV Mean	2.457	38.66	2.886	0.839	79.59

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

NOTES: This table presents the results of difference-in-difference regressions estimating the impact of AI model deployment on measures of productivity and agent performance. Observations are at the agent-month level, weighted by the number of chats associated with that agent-month. Post AI X Ever Treated measures the impact of AI model deployment after deployment on treated agents for resolutions per hour; average handle time; chats per hour; resolution rate; and net promoter score (NPS), an estimate of customer satisfaction. Regressions follow Equation 1 and include agent level, chat-year, and months of agent tenure fixed effects. Data are available at the agent-month level, and robust standard errors are clustered at the agent level.

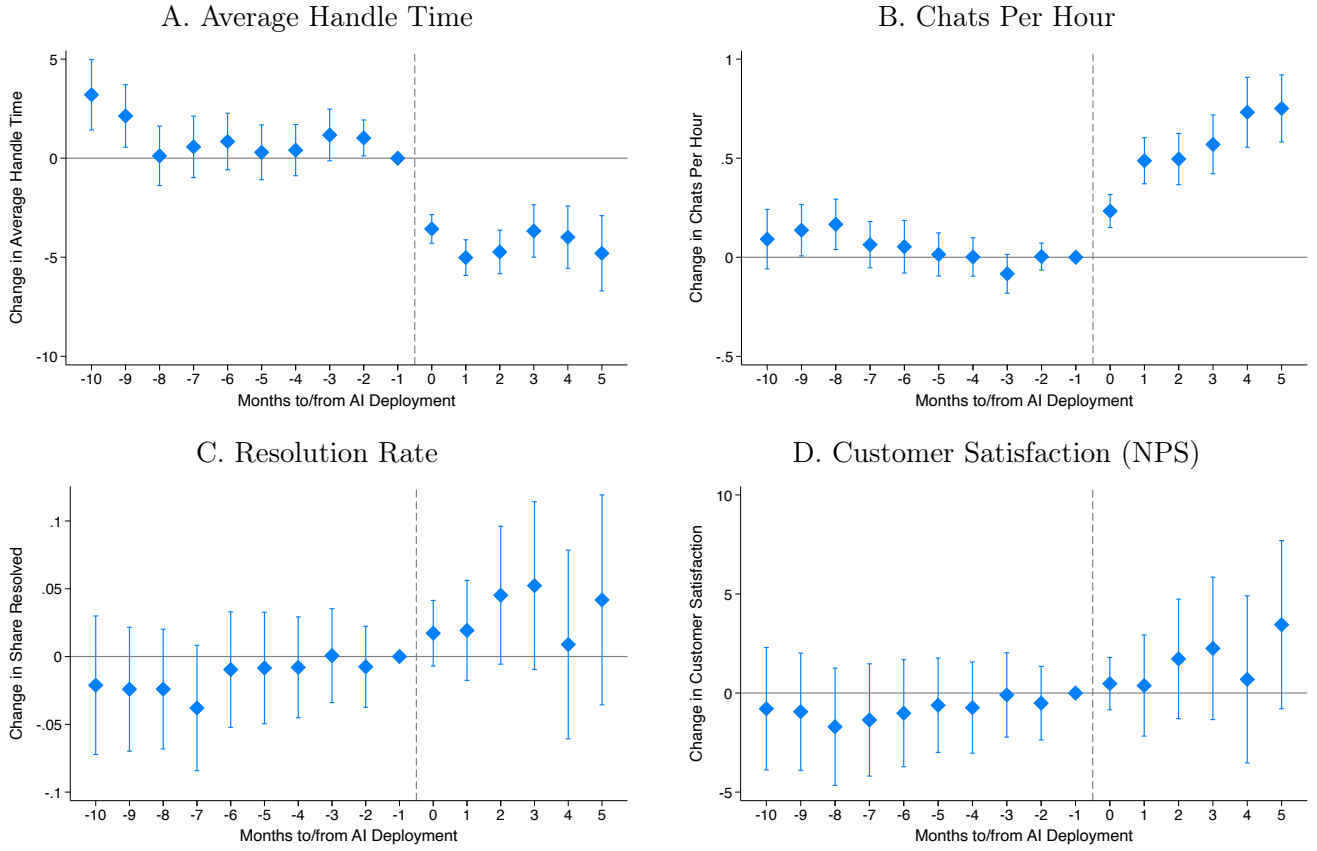
TABLE A.V

Main Effects: Productivity (Resolutions per Hour), Alternative Difference-in-Difference Estimators

	Point Estimate	Standard Error	Lower Bound 95% Confidence Interval	Upper Bound 95% Confidence Interval
TWFE-OLS	0.296	0.032	0.233	0.360
Borusyak-Jaravel-Spiess	0.576	0.070	0.438	0.714
Callaway-Sant’Anna	0.489	0.059	0.374	0.605
DeChaisemartin-D’Haultfeuille	0.219	0.042	0.137	0.302
Sun-Abraham	0.521	0.094	0.337	0.705

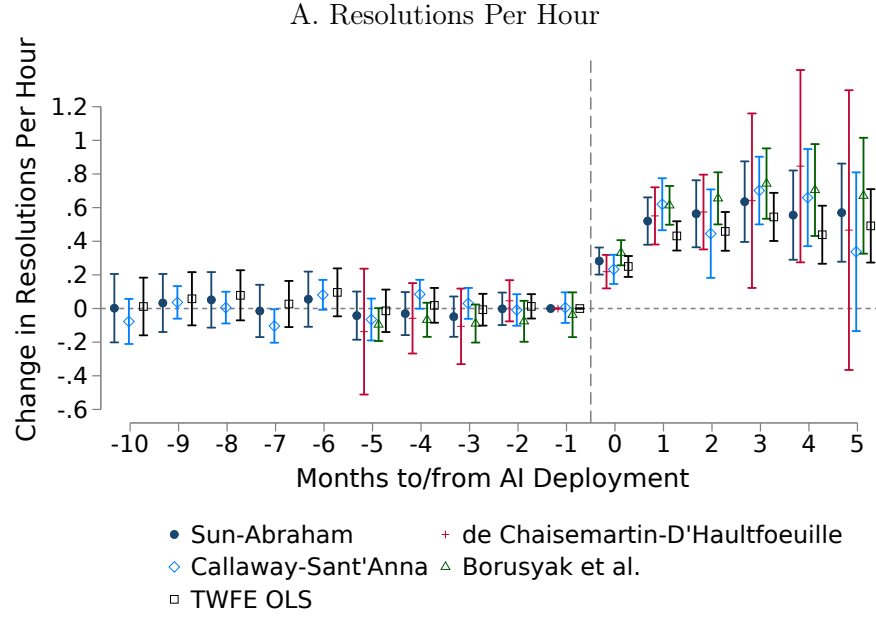
NOTES: This table shows the impact of AI model deployment on our main productivity outcome, resolutions per hour, using robust difference-in-differences estimators introduced in [Borusyak, Jaravel, and Spiess \(2024\)](#), [Callaway and Sant’Anna \(2021\)](#), [de Chaisemartin and D’Haultfeuille \(2020\)](#) and [Sun and Abraham \(2021\)](#). Regressions follow Equation 1 and include agent level, chat-year, and months of agent tenure fixed effects. Data are available at the agent-month level and standard errors are clustered at the agent level. Because of the number of post-treatment periods and high turnover of agents in our sample, we can only estimate five months of pre-period data using [Borusyak, Jaravel, and Spiess \(2024\)](#) and [de Chaisemartin and D’Haultfeuille \(2020\)](#).

FIGURE A.III
Event Studies, Additional Outcomes



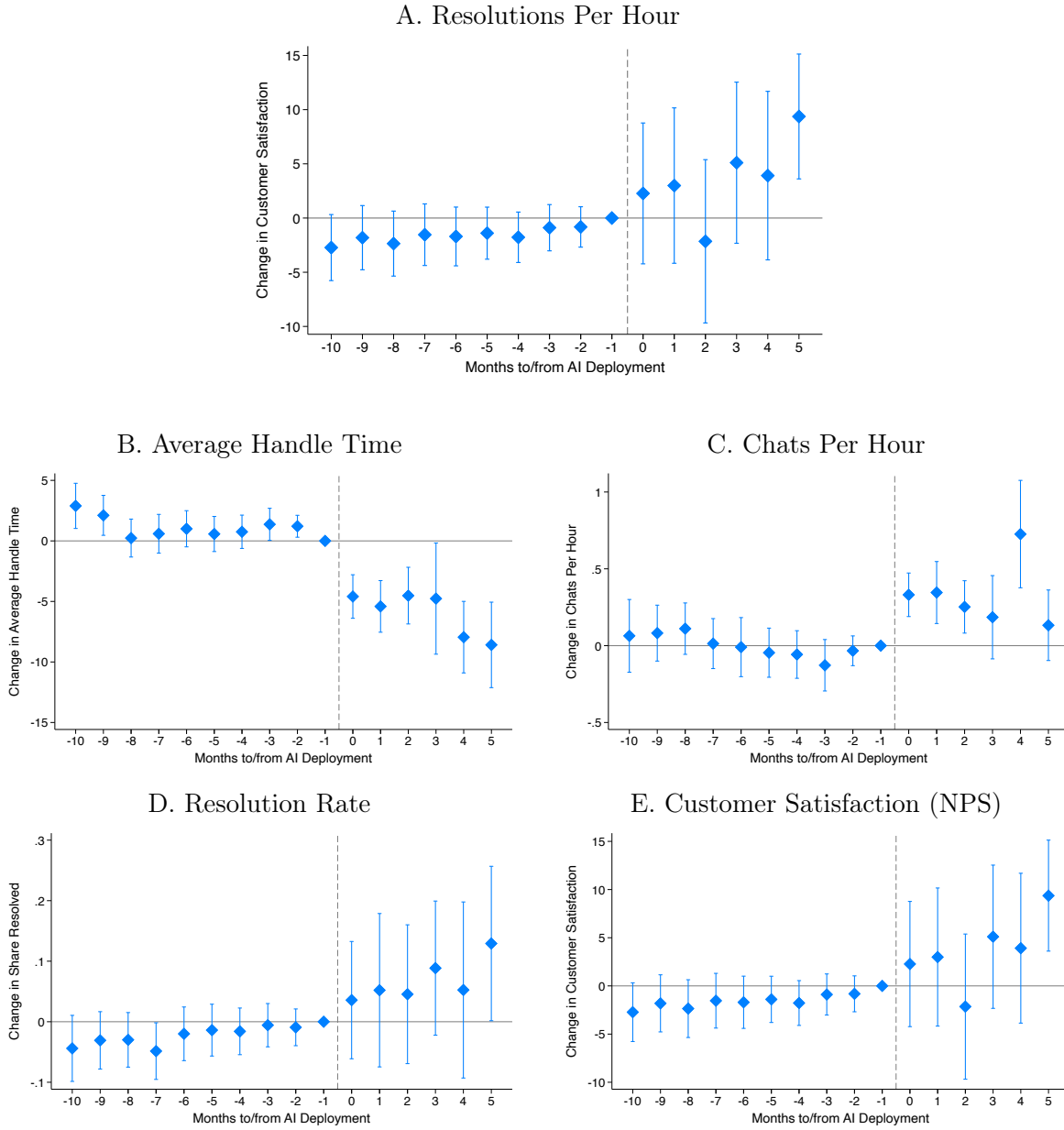
NOTES: These figures plot the coefficients and 95% confidence intervals from event study regressions of AI model deployment using the [Sun and Abraham \(2021\)](#) interaction weighted estimator. Panel A plots the average handle time or the average duration of each technical support chat. Panel B plots the number of chats an agent completes per hour, incorporating multitasking. Panel C plots the resolution rate, the share of chats successfully resolved; and Panel D plots net promoter score, which is an average of surveyed customer satisfaction. All specifications follow Equation 1 and include agent and chat year-month and months of agent tenure fixed effects. Data are at the agent-month level, and robust standard errors are clustered at the agent level.

FIGURE A.IV
Event Studies, Resolutions Per Hour



NOTES: This figure presents the effect of AI model deployment on our main productivity outcome, resolutions per hour, using a variety of robust dynamic difference-in-differences estimators introduced in Borusyak, Jaravel, and Spiess (2024), Callaway and Sant'Anna (2021), de Chaisemartin and D'Haultfoeuille (2020) and Sun and Abraham (2021) and a standard two-way fixed effects regression model. Regressions follow Equation 1 and include agent level, chat-year, and months of agent tenure fixed effects. Data are available at the agent-month level, and robust standard errors are clustered at the agent level. Because of the number of post-treatment periods and high turnover of agents in our sample, we can only estimate five months of pre-period data using Borusyak, Jaravel, and Spiess (2024) and de Chaisemartin and D'Haultfoeuille (2020).

FIGURE A.V
Randomized Control Trial Analysis



NOTES: This figures plots event studies focusing on the 22 agents who were on-boarded as part of the pilot randomized control trial (RCT). Because we do not have information on the specific agents (there were around 25) selected to be part of the control group, we compare the 22 RCT treated agents with observations for all pre-treatment agents, controlling for our usual agent, chat year-month, and months of agent tenure fixed effects in agent-month level regressions following Equation 1. Robust standard errors are clustered at the agent level.

C. IMPACTS BY AGENT SKILL AND TENURE

TABLE A.VI
Heterogeneity of AI Impact by Skill and Tenure, Resolutions per Hour

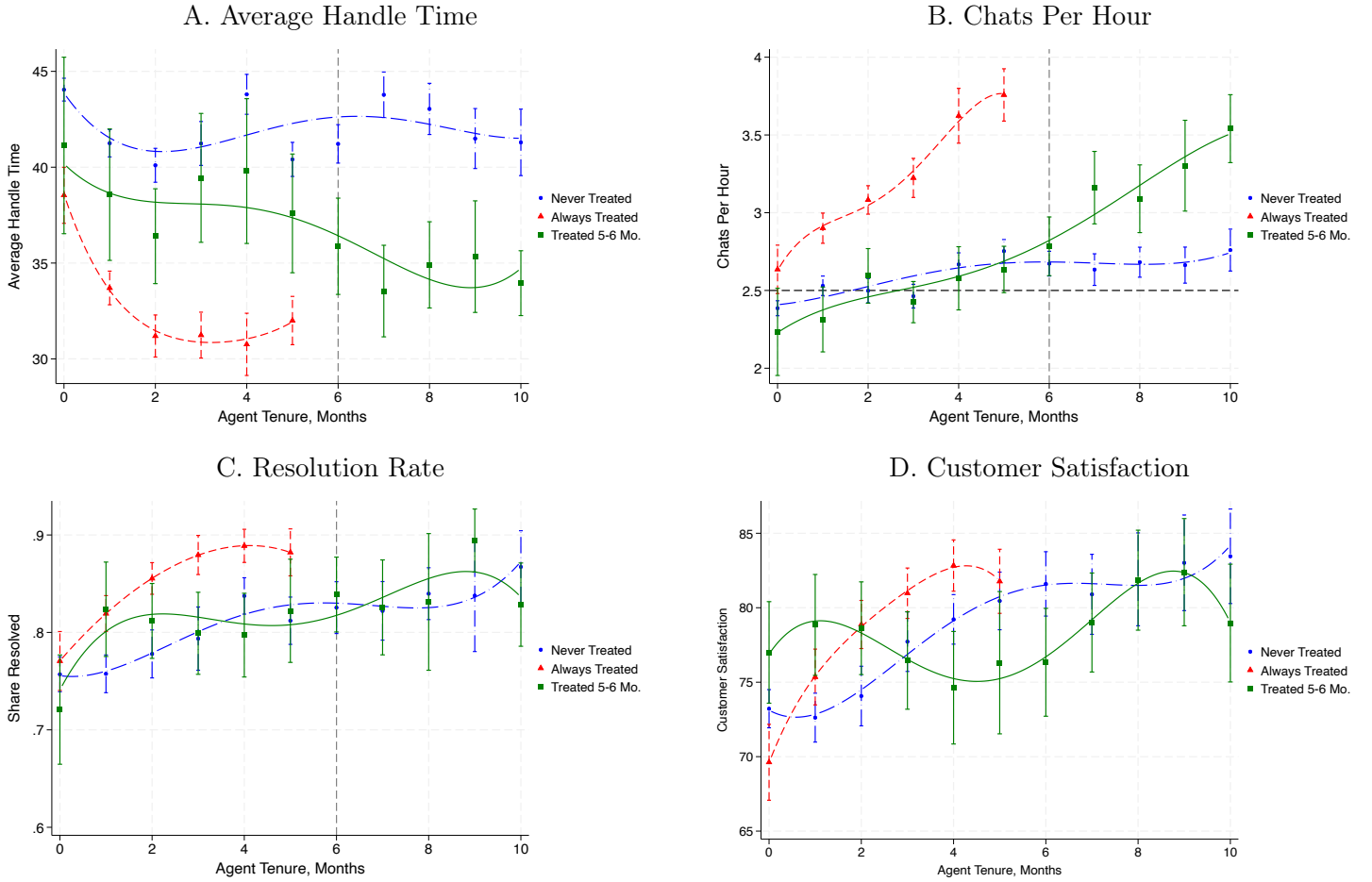
	(1) By Skill at AI	(2) By Tenure at AI
Q1 (Lowest Skill)	0.527*** (0.049)	
Q2	0.315*** (0.056)	
Q3	0.302*** (0.043)	
Q4	0.261*** (0.055)	
Q5 (Highest Skill)	0.015 (0.065)	
< 1 Mos.)		0.707*** (0.067)
1-2 Mos.		0.388*** (0.042)
3-6 Mos.		0.361*** (0.067)
7-12 Mos.		0.337*** (0.054)
> 12 Mos.)		0.028 (0.059)
Year Month FE	Yes	Yes
Agent FE	Yes	Yes
Other FE	Tenure	Skill at AI
DV Mean	2.176	2.284
Observations	12,295	8,148

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

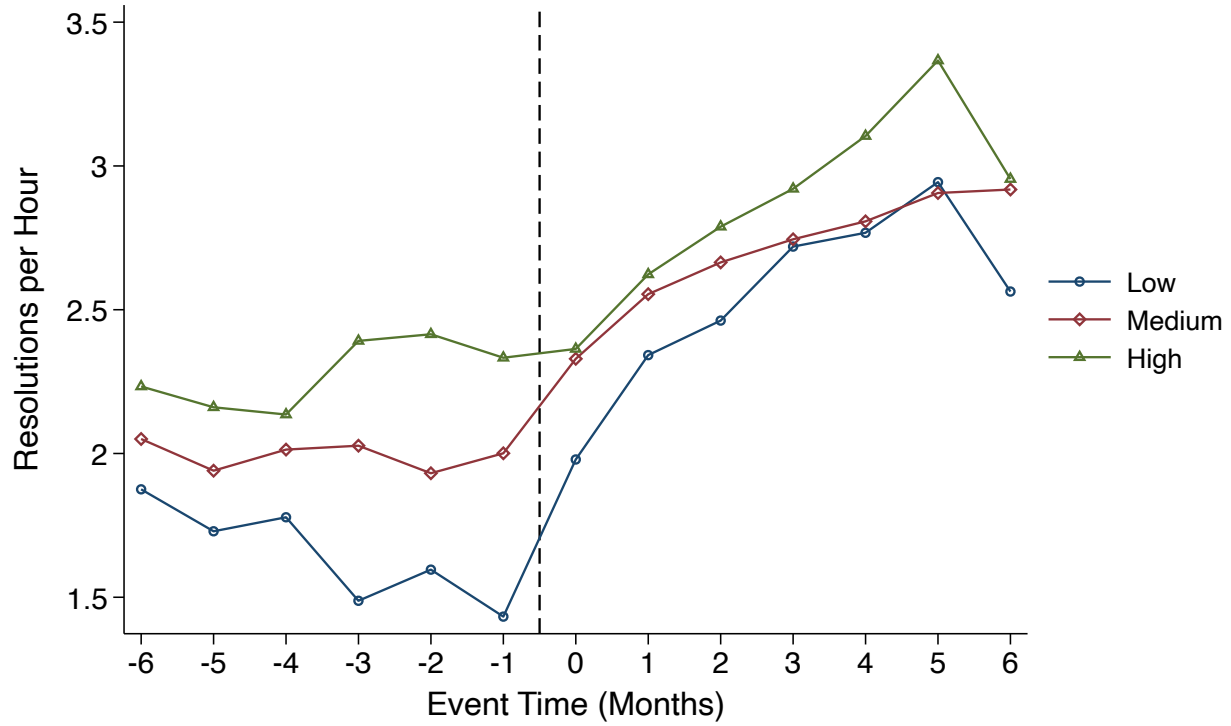
NOTES: This table presents the results of difference-in-difference regressions estimating the impact of AI model deployment on resolutions per hour. Column 1 estimates the impact of AI access by worker skill at AI, including agent, year-month, and months of agent tenure fixed effects. Column 2 estimates the effects by worker tenure at AI deployment, including agent, year-month, and agent skill at AI deployment fixed effects. Observations are at the agent-month level, and all standard errors are clustered at the agent level.

FIGURE A.VI
Experience Curves by Deployment Cohort, Additional Outcomes



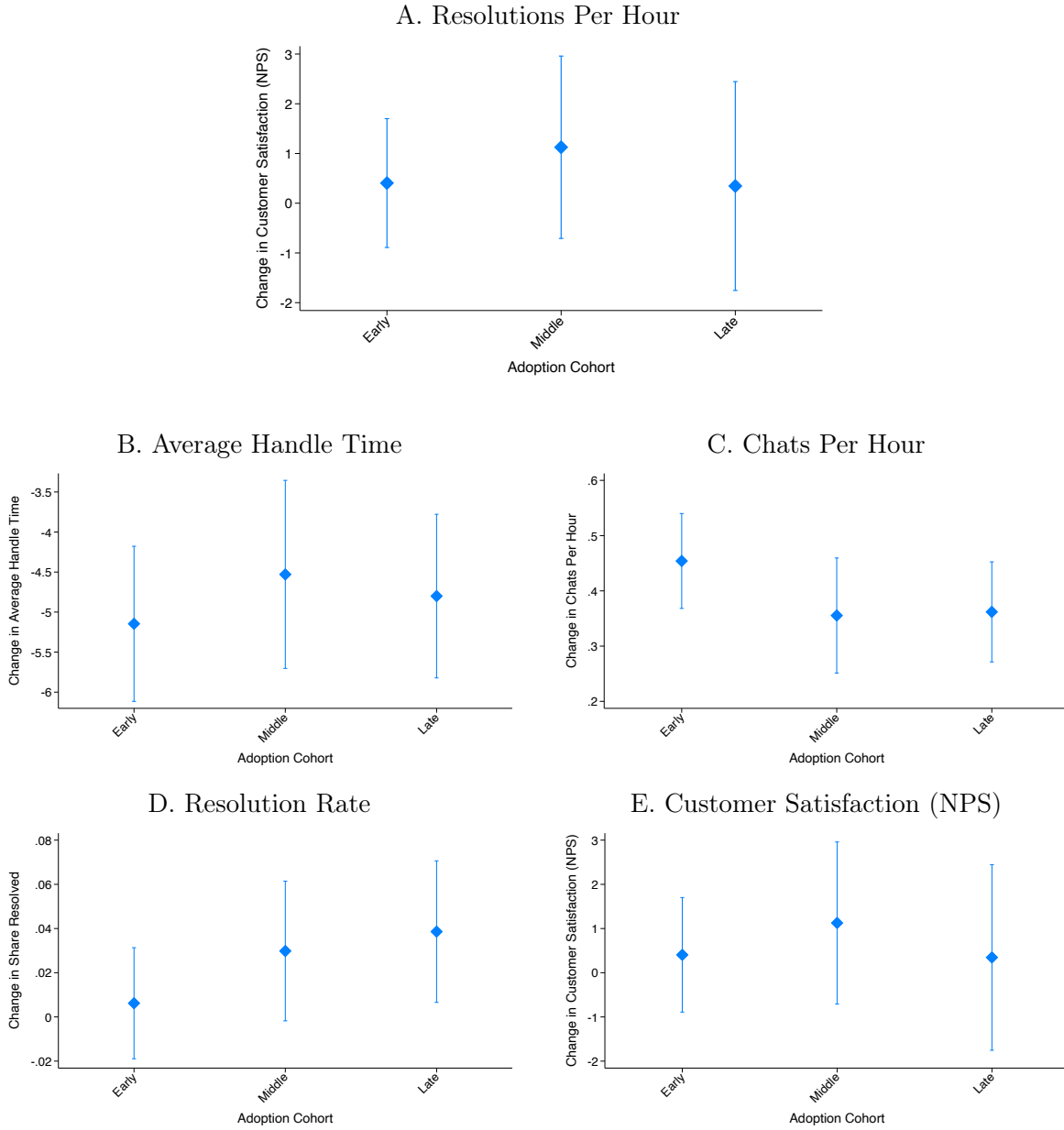
NOTES: These figures plot the experience curves of three groups of agents over their tenure, the x-axis, against five measures of productivity and performance. The short-dashed lines plot the performance of always-treated agents, those who start work in their first month with the AI and always have access to the AI suggestions. The long-dashed lines plot agents who are never treated. The solid lines plot agents who spend their first four months of work without the AI model and gain access to the AI during their fifth month on the job. All panels include 95% confidence intervals, and observations are at the agent-month level.

FIGURE A.VII
Resolutions per Hour over Time



NOTES: This graph depicts the evolution of average resolutions per hour for agents following the implementation of AI assistance. The graph segments agents into three groups based on their skill level at the time of AI deployment. The triangle-symbol line represents the highest-performing third of agents, those in the top tercile of the skill index. The diamond-symbol line illustrates the progress of agents in the middle tercile, while the circle-symbol line tracks those in the bottom tercile, representing the lowest-skilled third at the time of treatment. Agents are categorized based on their skill index at the time of AI implementation. For details on the skill index construction, refer to Appendix Section J.B.

FIGURE A.VIII
Productivity Impacts by Adoption Cohort



NOTES: These figures plot the treatment effect of AI assistance on various outcomes for workers who received access to the AI model in the early, middle, or later period of the rollout. Because the AI model is periodically updated with new data and pushed to all on-boarded workers no more frequently than once a month, we focus on productivity outcomes in the first month after AI adoption in order to compare workers using earlier and later versions of the model. Data are at the agent-month; regression specification follows Equation 1; and we include chat year-month, agent, and months of tenure fixed effects. Robust standard errors are clustered at the agent level.

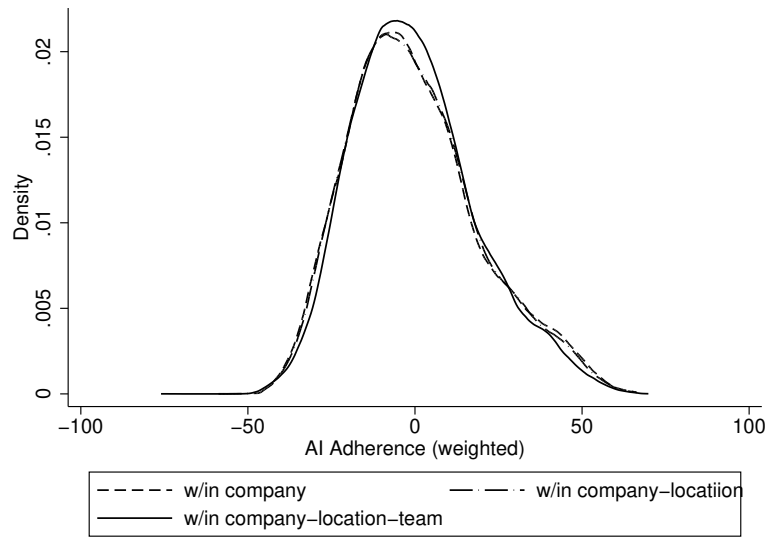
D. ADHERENCE TO AI SUGGESTIONS

TABLE A.VII
Heterogeneity of AI Impact by Initial AI Adherence, Resolutions per Hour

	(1) By Adherence at AI
Q1 (Lowest Adherence)	0.213*** (0.043)
Q2	0.233*** (0.042)
Q3	0.293*** (0.042)
Q4	0.309*** (0.042)
Q5 (Highest Adherence)	0.432*** (0.043)
Year Month FE	Yes
Agent FE	Yes
Agent Tenure FE	Yes
DV Mean	2.176
Observations	12,295
Standard errors in parentheses	
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

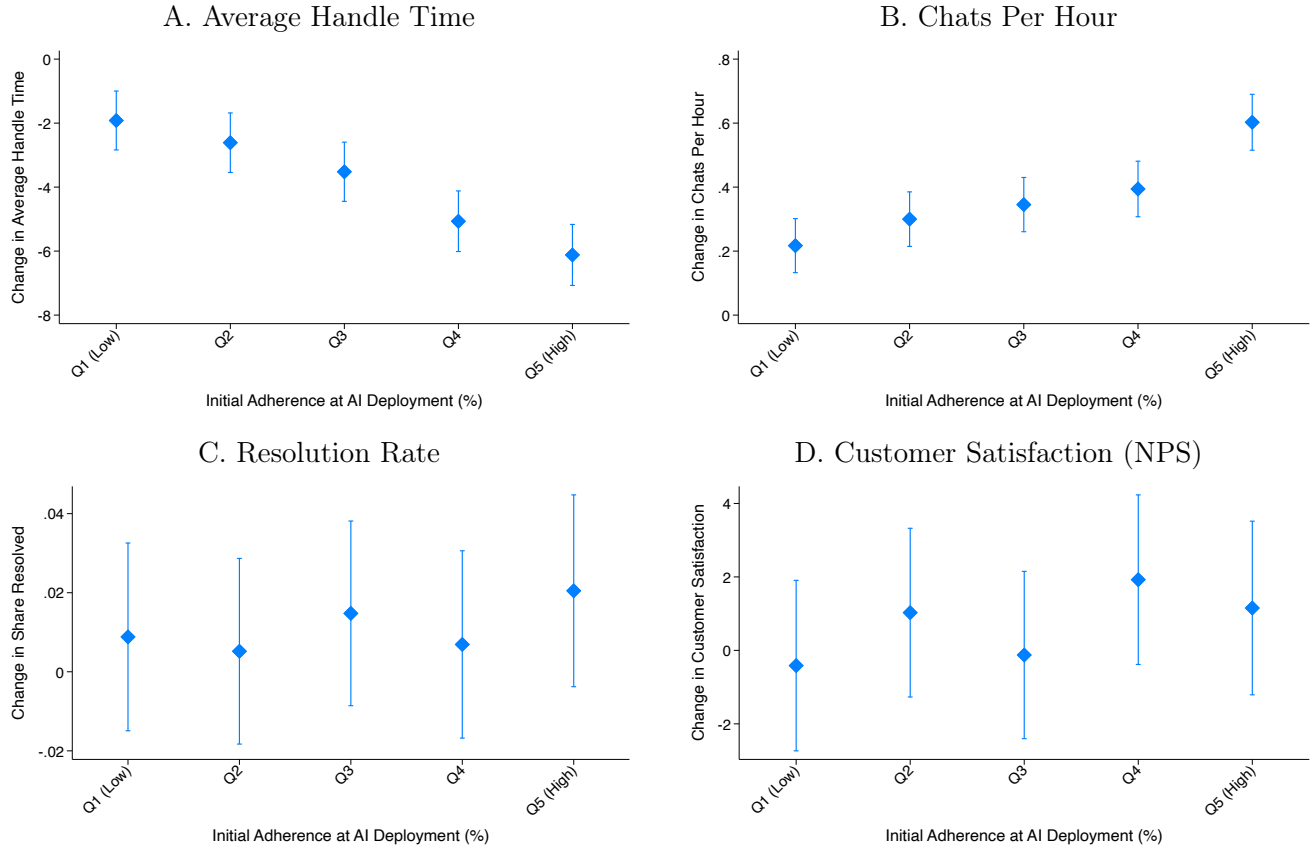
NOTES: This table presents the results of difference-in-difference regressions estimating the impact of AI model deployment on resolutions per hour by initial adherence. Column 1 estimates the impact of AI access by initial adherence quintile, including agent, year-month, and months of agent tenure fixed effects. Observations are at the agent-month level, and all standard errors are clustered at the agent level. We include chat year-month, agent, and months of agent tenure fixed effects.

FIGURE A.IX
Variation in AI Adherence



NOTES: This figure plots the distribution of AI adherence, averaged at the agent-month level, weighted by the log of the number of AI recommendations for that agent-month. “Within company” refers to the overall distribution of adherence rates, adjusted by the company of employment. “Within company-location” plots residual adherence after adjusting for company and location fixed effects; and “within company-location-team” plots residuals adjusted for company, location, and team fixed effects.

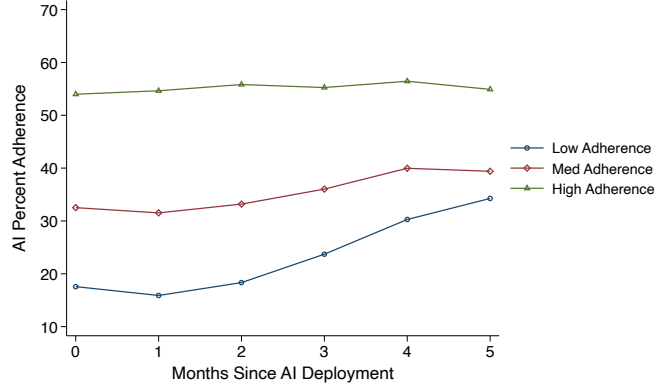
FIGURE A.X
Heterogeneity of AI Impact by Initial AI Adherence, Additional Outcomes



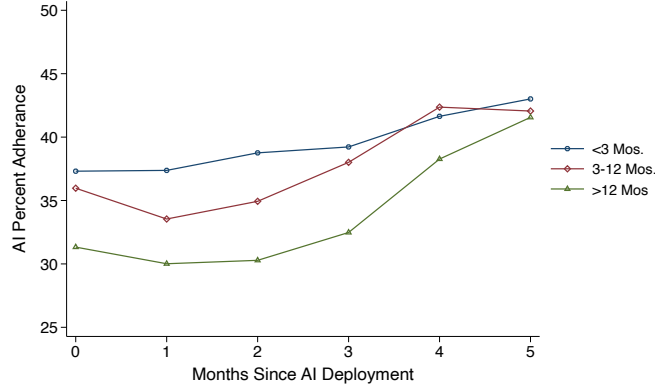
NOTES: These figures plot the impact of AI model deployment on additional measures of performance by quintile of initial adherence, the share of AI recommendations followed in the first month of treatment. Panel A plots the average handle time or the average duration of each technical support chat. Panel B graphs chats per hour, the number of chats an agent can handle per hour (including working on multiple chats simultaneously). Panel C plots the resolution rate, the share of chats successfully resolved. Panel D plots NPS, or net promoter score, an average of surveyed customer satisfaction. Data are at the agent level, and all regressions include agent and chat year-month, months of agent tenure, with more details in Appendix section J.C.

FIGURE A.XI
AI Adherence over Time

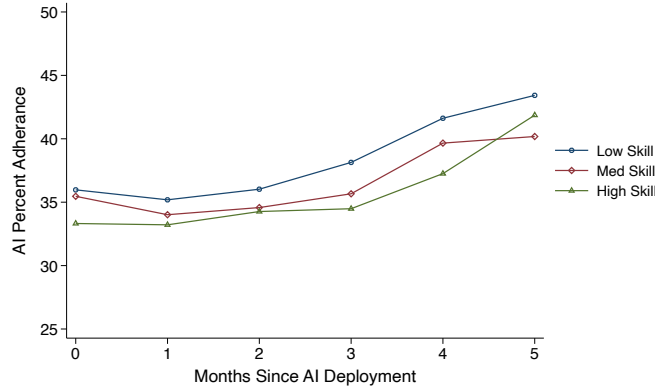
A. By Adherence at AI Model Deployment



B. By Agent Tenure at AI Model Deployment



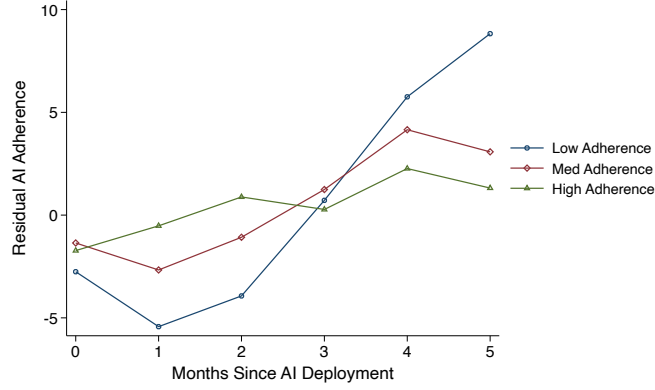
C. By Agent Skill at AI Model Deployment



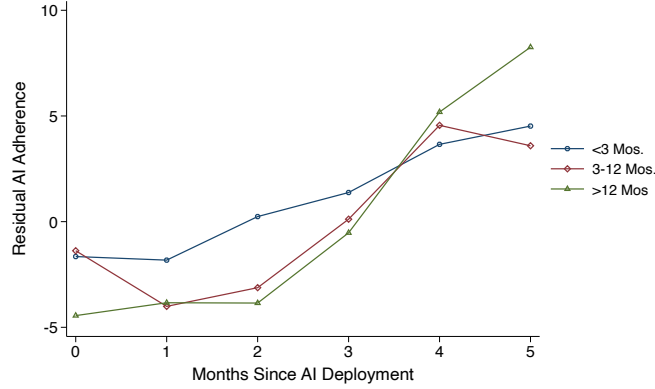
NOTES: This figure plots the share of AI suggestions followed by agents as a function of the number of months each agent has had access to the AI model. In Panel A, we divide agents into terciles based on their adherence to AI suggestions in the first month. In Panel B, we divide agents into groups based on their tenure at the firm at the time of AI model deployment. In Panel C, we divide workers into terciles of pre-deployment productivity as defined by our skill index.

FIGURE A.XII
Within-Agent AI Adherence over Time

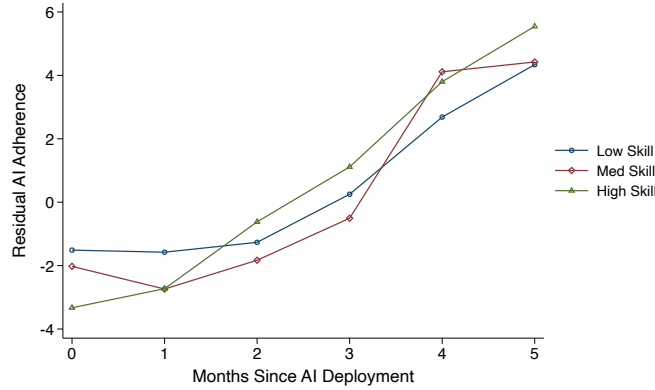
A. By Adherence at AI Model Deployment



B. By Agent Tenure at AI Model Deployment



C. By Agent Skill at AI Model Deployment



NOTES: This figure plots the residualized percentage of AI suggestions followed by agents as a function of the number of months each agent has had access to the AI model, after controlling for agent level fixed effects. In Panel A, we divide agents into terciles based on their adherence to AI suggestions in the first month. In Panel B, we divide agents into groups based on their tenure at the firm at the time of AI model deployment. In Panel C, we divide workers into terciles of pre-deployment productivity as defined by our skill index.

E. WORKER LEARNING

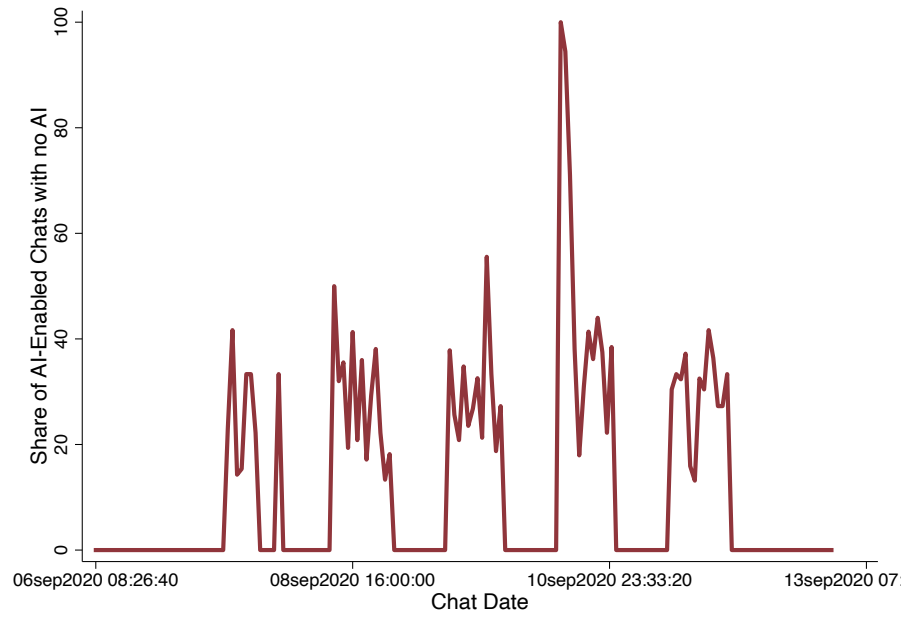
TABLE A.VIII
Chat Duration during AI System Outages

	(1) Non Outage	(2) AI Outages	(3) Outages High Receptivity	(4) Outages Low Receptivity
lead10	0.684 (0.550)	2.085*** (0.581)	3.391*** (0.880)	0.284 (1.590)
lead9	0.549 (0.529)	2.126*** (0.551)	4.056*** (0.890)	1.003 (1.388)
lead8	-0.222 (0.479)	1.410*** (0.504)	3.239*** (0.771)	0.340 (1.299)
lead7	-0.302 (0.470)	1.267*** (0.486)	2.354*** (0.798)	1.025 (1.141)
lead6	0.874* (0.463)	2.204*** (0.468)	4.081*** (0.786)	2.210** (1.079)
lead5	0.439 (0.389)	1.444*** (0.378)	2.953*** (0.591)	1.736* (0.885)
lead4	0.415 (0.336)	1.052*** (0.321)	2.222*** (0.471)	1.258* (0.703)
lead3	0.135 (0.305)	0.343 (0.297)	1.314*** (0.482)	0.218 (0.627)
lead2	-0.146 (0.248)	-0.209 (0.247)	0.723* (0.411)	-0.869* (0.469)
lag0	-4.105*** (0.270)	-3.471 (3.713)	-5.236 (3.732)	9.410*** (3.564)
lag1	-4.523*** (0.313)	-2.420 (3.242)	-6.699* (4.052)	0.000 (.)
lag2	-4.785*** (0.335)	0.767 (3.112)	-0.201 (3.858)	0.000 (.)
lag3	-4.424*** (0.333)	-2.851 (2.663)	-2.677 (3.915)	6.005 (4.099)
lag4	-4.225*** (0.350)	-2.423* (1.442)	-4.011** (2.015)	4.851 (3.663)
lag5	-4.084*** (0.380)	0.838 (1.744)	0.695 (2.042)	7.871 (5.904)
lag6	-4.300*** (0.388)	-7.799*** (1.836)	-8.574*** (2.168)	-2.303 (3.167)
lag7	-4.445*** (0.393)	-8.494*** (1.635)	-9.532*** (1.687)	-2.161 (3.209)
lag8	-3.804*** (0.411)	-4.672* (2.647)	-2.163 (2.471)	-8.059*** (3.023)
lag9	-3.628*** (0.440)	-8.705*** (1.660)	-8.842*** (1.930)	0.318 (5.166)
lag10	-3.609*** (0.501)	-4.386** (1.720)	-4.486** (2.040)	-0.233 (4.273)
lag11	-3.305*** (0.486)	-2.779** (1.380)	-6.857** (3.252)	0.539 (4.003)
lag12	-3.479*** (0.517)	-1.966 (3.003)	-4.630 (4.248)	-8.939** (3.510)
Year Month FE	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes
DV Mean	41.982	41.982	41.982	41.982
R-squared	.12	.104	.113	.083
Observations	2,969,371	1,829,527	1,220,852	323,810

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

NOTES: This table shows the event study coefficients corresponding to the figures in VII. Column 1 shows the impact of AI access on call duration during non-outage periods. Column 2 shows the impact when the AI system is experiencing an outage. Column 3 shows the effects on agents who are in the top tercile of AI adherence, while Column 4 shows the impacts on agents in the lowest tercile of adherence. Observations for these OLS regressions are at the agent-chat level, and robust standard errors are clustered at the agent level. All specifications include agent, chat year-month, and months of agent tenure fixed effects.

FIGURE A.XIII
Sample AI Outage



NOTES: This figure plots the share of post-treatment chats with no AI suggestions during a period of a documented software outage.

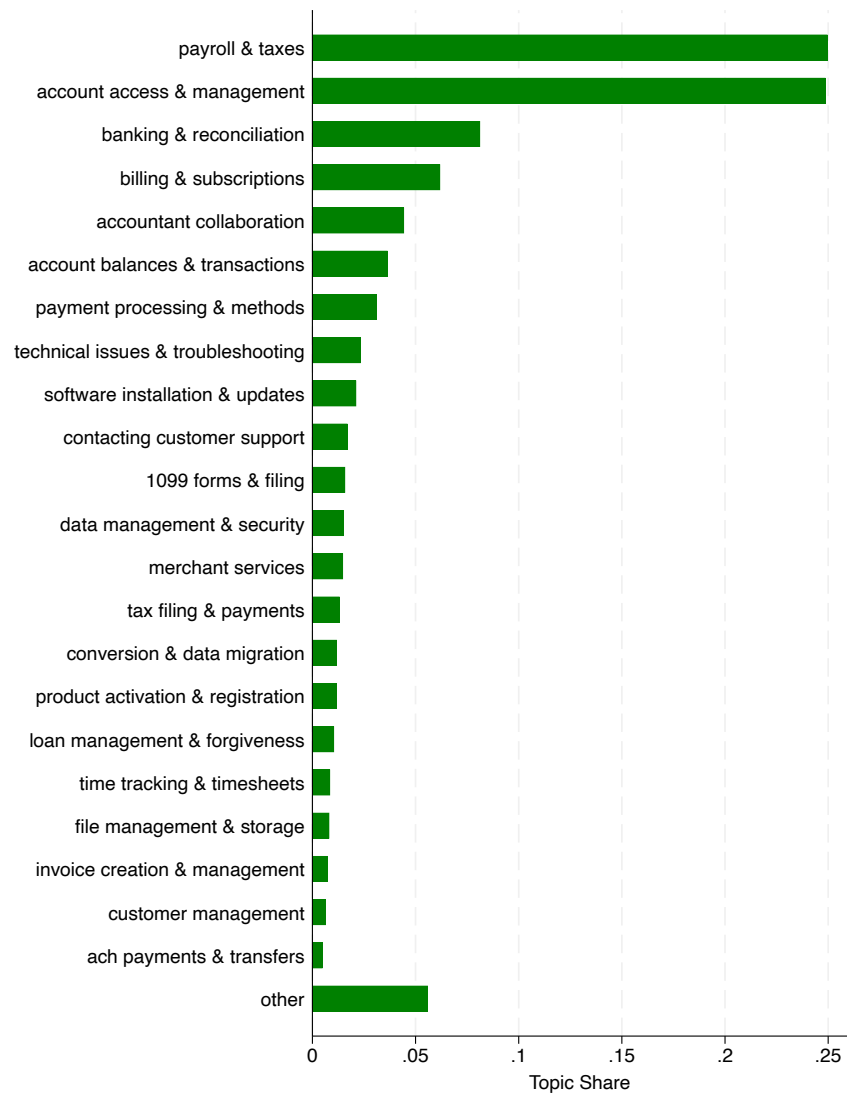
F. TOPIC HETEROGENEITY

TABLE A.IX
Impact of AI on Chat Duration, by Topic Commonality

VARIABLES	(1) Call Duration Aggregate Problem Frequency	(2) Call Duration Within-Agent Frequency
Q1 (Most Common Overall)	-0.107*** (0.00717)	
Q2	-0.115*** (0.00724)	
Q3	-0.136*** (0.00742)	
Q4 (Least Common Overall)	-0.118*** (0.00842)	
Q1 (Most Common within Agent)		-0.100*** (0.00927)
Q2		-0.103*** (0.00740)
Q3		-0.119*** (0.00700)
Q4 (Least Common within Agent)		-0.142*** (0.00708)
Observations	2,089,995	2,089,995
R-squared	0.120	0.122
Year Month FE	Yes	Yes
Agent FE	Yes	Yes
Agent Tenure FE	Yes	Yes
DV Mean	0.916	0.915
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.10		

NOTES: This table shows the impact of AI model deployment on call duration by frequency of the chat topic. Regression includes controls for chat year-month, agent, and months of agent tenure. Data are at the chat level, and robust standard errors are clustered at the agent level. Appendix section J.B describes construction of topics and the regression specifications.

FIGURE A.XIV
Distribution of Conversational Topics



NOTES: This figure shows the distribution of chats by common topics in our data.

G. LANGUAGE FLUENCY

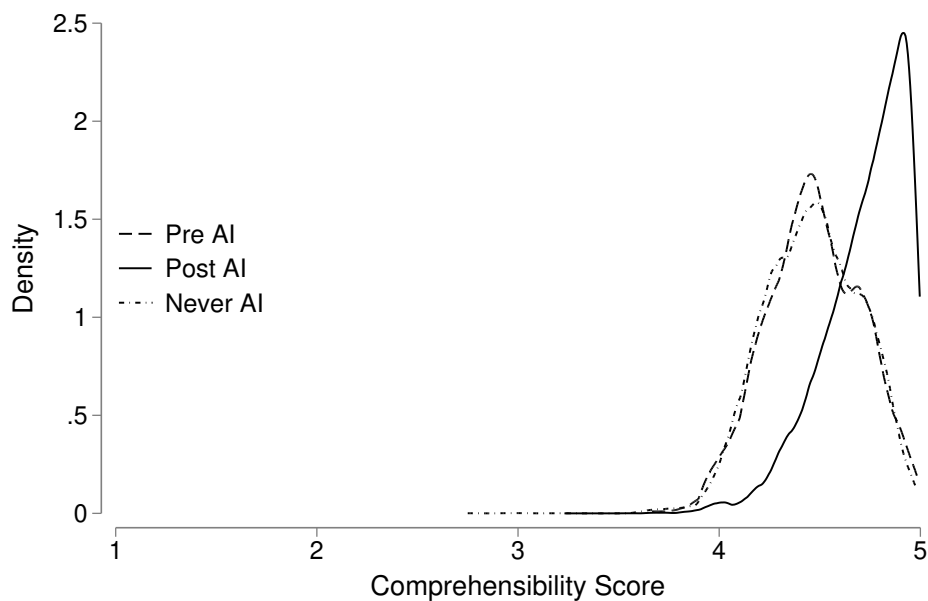
TABLE A.X
AI Impact on Language Fluency and Comprehensibility

VARIABLES	(1) Native Fluency	(2) Comprehensibility	(3) Native Fluency	(4) Comprehensibility
Post AI X Ever Treated	0.250*** (0.00851)	0.241*** (0.00767)		
Post AI X Ever Treated X US = 1			0.159*** (0.0225)	0.134*** (0.0208)
Post AI X Ever Treated X Philippines = 1			0.251*** (0.00684)	0.234*** (0.00672)
Observations	12,772	12,772	11,271	11,271
R-squared	0.791	0.754	0.733	0.751
Year Month FE	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes
DV Mean	4.564	4.597	4.592	4.608
Robust standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.10				

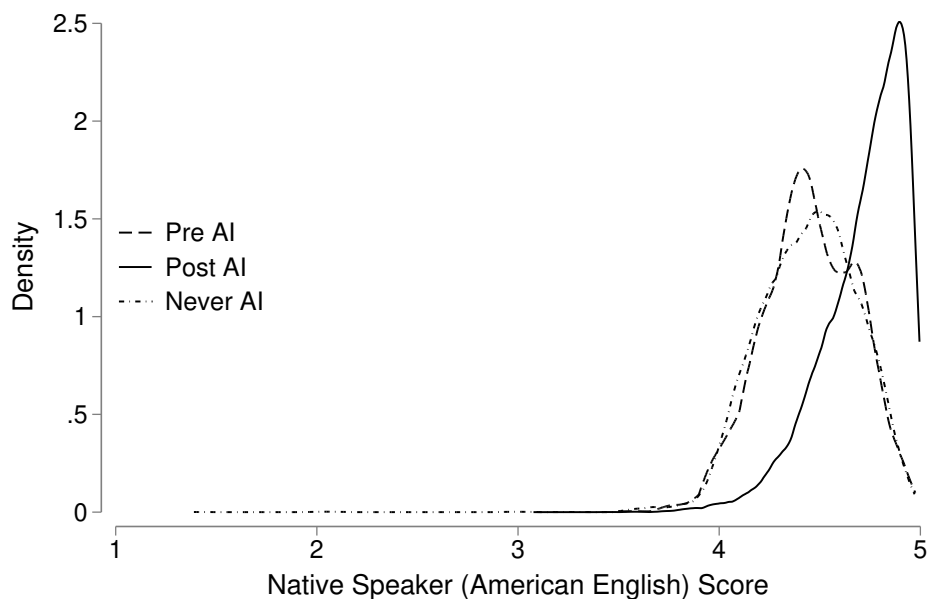
NOTES: This table presents the results of difference-in-difference regressions estimating the impact of AI model deployment on native fluency and comprehensibility. Observations are aggregated to the agent-month level, and regressions include chat year-month, agent, and months of agent tenure fixed effects. In Columns 1 and 2, robust standard errors are clustered at the agent level, while at the agent location level in Columns 3 and 4. Appendix Section J.B describes construction of the language scores.

FIGURE A.XV
Distribution of Language Skills

A. Comprehensibility Score



B. Native Fluency Score

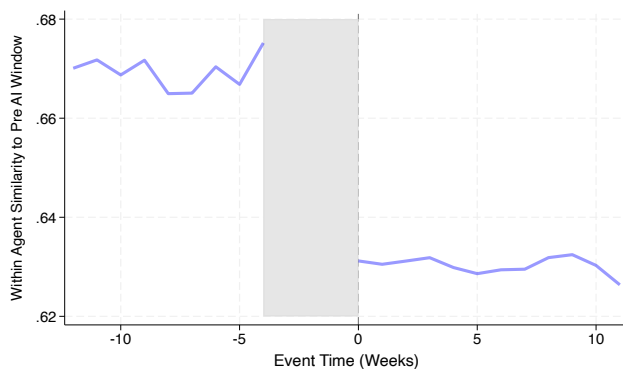


NOTES: These figures show the distributions of comprehensibility and native fluency scores. We split this sample into agent-month observations for agents who eventually receive access to the AI system before deployment (“Pre AI”), after deployment (“Post AI”), and for agent-months associated with agents who never receive access (“Never AI”). Appendix Section J.B describes construction of the language scores.

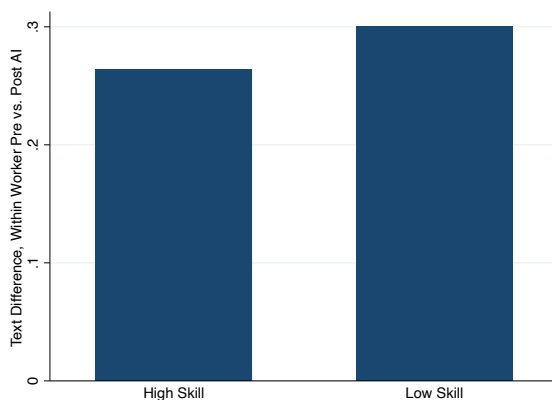
H. TEXTUAL CONVERGENCE

FIGURE A.XVI
Within-Agent Textual Analysis

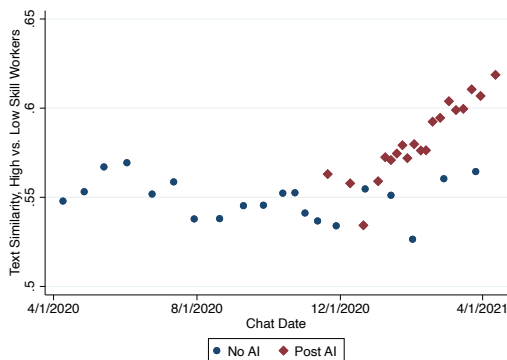
A. Within-Person Textual Similarity to Month Prior to AI



B. Within-Person Textual Change, Low vs. High Skill



C. Text Similarity between Low-Skill and High-Skill Workers, Pre and Post AI



NOTES: Panel A plots the average similarity between an agent's chats each week and a comparison group of their conversations in the month prior to AI deployment (the gray section). Panel B plots the average difference between an agent's pre-AI corpus of chat messages and that same agent's post-AI corpus, controlling for year-month and agent tenure. The first bar represents the average pre-post text difference for agents in the highest quintile of pre-AI skill, and the second bar represents those in the bottom quintile. Panel C plots the average text similarity between the top and bottom quintile of agents. The circular points plot the similarity for never treated or pre-treatment agents; the square points plot the similarity for agents with access to the AI model. For agents in the treatment group, we define agent skill at AI model deployment.

I. EXPERIENCE OF WORK

TABLE A.XI
Attrition

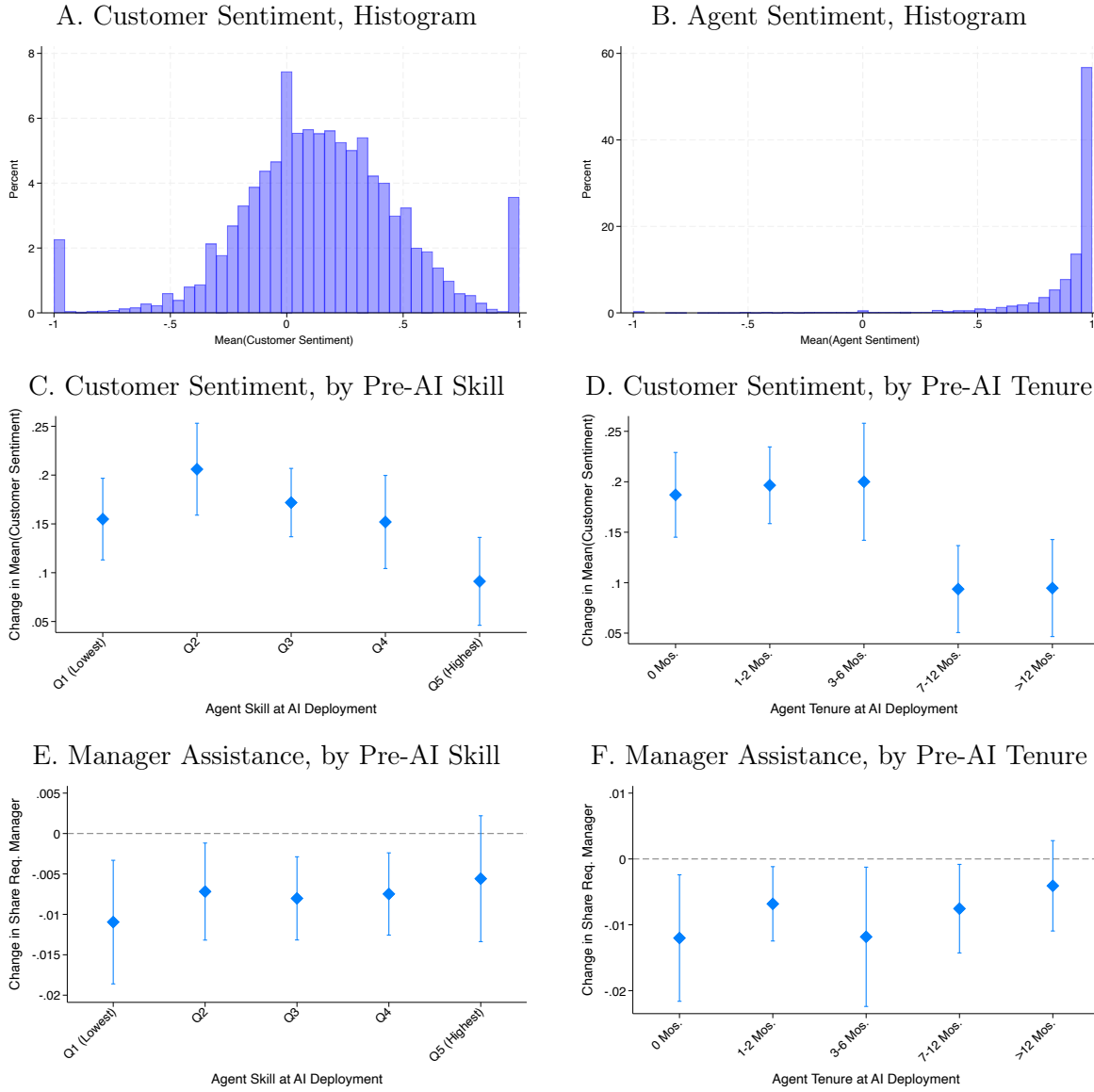
VARIABLES	(1) Leaves this Month	(2) Leaves this Month	(3) Leaves this Month
Post AI X Ever Treated	-0.0868*** (0.0130)		
0 Mos.		-0.0952*** (0.0162)	
1-2 Mos.		-0.121*** (0.0167)	
3-6 Mos.		-0.0850*** (0.0190)	
7-12 Mos.		-0.0803*** (0.0165)	
>12 Mos.		-0.0306 (0.0234)	
Q1 (Low Skill)			-0.0655*** (0.0148)
Q2			-0.0864*** (0.0134)
Q3			-0.0741*** (0.0124)
Q4			-0.0936*** (0.0146)
Q5 (High Skill)			-0.0531*** (0.0145)
Observations	17,902	17,902	17,902
R-squared	0.206	0.206	0.206
Year Month FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes
Agent Company FE	Yes	Yes	Yes
DV Mean	0.288	0.288	0.288

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

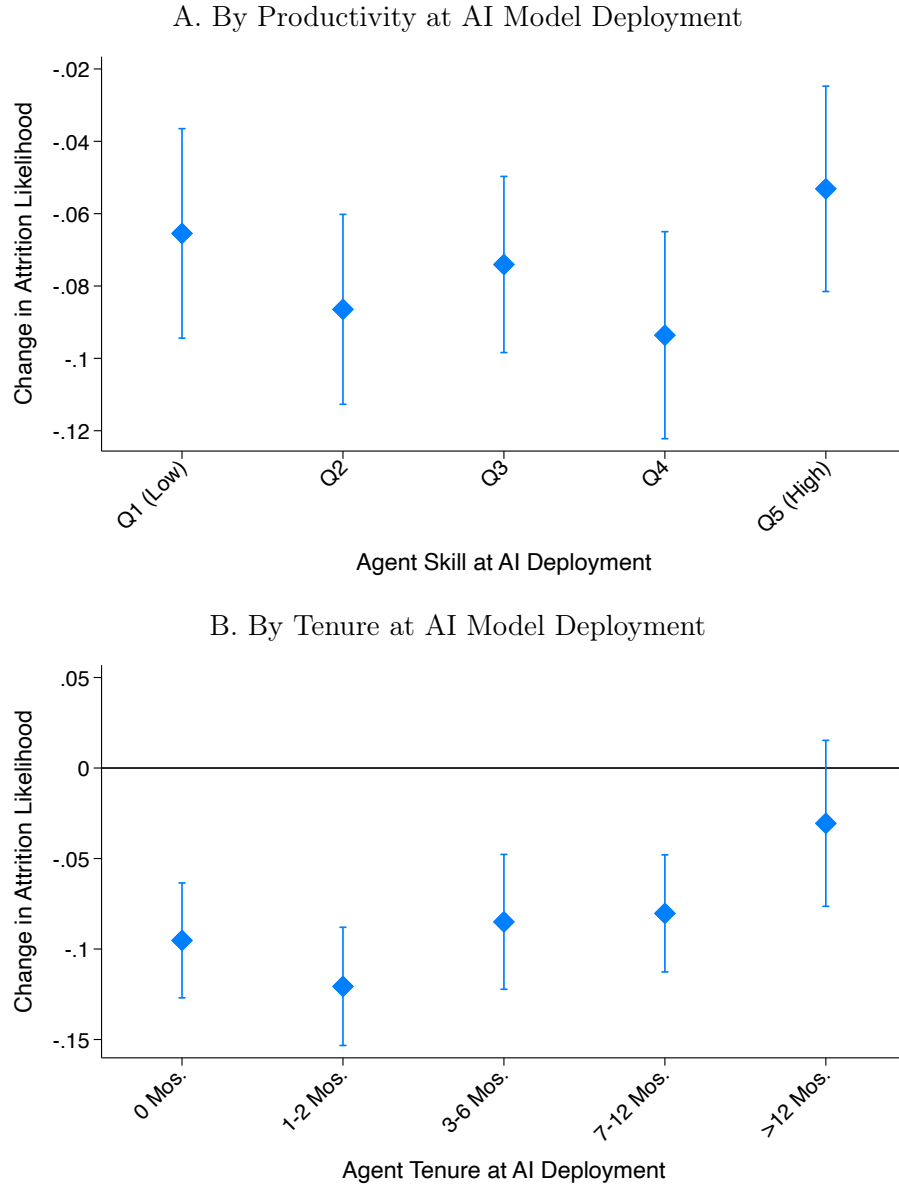
NOTES: This table presents the results of difference-in-difference regressions estimating the impact of AI model deployment on our main measure of productivity, resolutions per hour, the number of technical support problems resolved by an agent per hour (resolutions/hour). Post AI X Ever Treated captures the impact of AI model deployment on resolutions per hour. Column 1 includes agent geographic location and year-by-month fixed effects. Columns 2 and 3 include agent-level fixed effects. Column 3, our preferred specification described by Equation 1, also includes fixed effects that control for months of agent tenure. Observations for this regression are at the agent-month level, and all standard errors are clustered at the agent level. Section IV.A describes the AI rollout procedure.

FIGURE A.XVII
Experience of Work



NOTES: Each panel of this figure illustrates the impact of AI model deployment on aspects of the experience of work. Panel A shows average customer sentiment, while Panel B shows average agent sentiment. Sentiment is measured using SiEBERT, a fine-tuned checkpoint of a RoBERTa, an English language transformer model. Panel C plots the impacts of AI on customer sentiment by agent ex-ante productivity, and Panel D plots the effects by agent tenure when the AI is deployed. Panels E and F show the effects of AI on customer requests for manager assistance, by pre-AI agent skill and by pre-AI agent tenure. Observations for the regression are at the agent-month level; robust standard errors are clustered at the agent level; and regressions include controls for agent, chat year-month, and months-of-tenure.

FIGURE A.XVIII
Impact of AI Model Deployment on Worker Attrition



NOTES: This figure presents the results of the impact of AI model deployment on workers' likelihood of attrition. Panel A plots the same impact by agent skill index at AI model deployment. Panel B graphs the effects of AI assistance on attrition by agent tenure at AI model deployment. All specifications include chat year and month fixed effects, as well as agent location, company, and agent tenure. Observations for these regressions, detailed in J.C, are at the agent-month, and all robust standard errors are clustered at the agent level. Regression results are in Appendix Table A.XI.

J. DATA APPENDIX

J.A. SAMPLE CONSTRUCTION

We begin with 3,006,395 chat conversations conducted by agents employed at our data firm over the period between September 2019 and June 2021. Each chat includes the number of messages in the conversation, the start and end times, and identifiers associated with the conversation and agent. We drop chats with only one agent or one customer message to avoid capturing interactions without meaningful content.

We merge the chat data with a set of internal company datasets and datasets from our AI firm that allow us to track agent information, conversation outcomes, and AI model output. To do this, we use the system-generated chat identifier to merge chat-level information across database systems. We also merge our conversation-level data into a message-level dataset that includes the text of each message and message-level AI output. At the conversation level, the difference between the chat start and end times gives us the chat duration, which is available for each call. We drop the small number of chats that are missing start or end times, not associated with an agent identifier, or missing a chat identifier. For the small number of chats that remain “open” for days because the customer and agents forget to close the chat, we winsorize call duration at the 99th percentile.

We aggregate our chat-level dataset to the agent-month level and merge it with agent data. This information includes the agents’ employer, their location, their manager/team, their tenure at the firm, and the date treated agents are on-boarded onto the AI assistant (defined as the date the agent’s account on the AI system is created). Some employees work flexible schedules, including part-time or seasonal roles. As a result, we measure performance metrics only for active periods, and agents’ tenure increases solely during months when they are actively handling customer chats.

J.B. CONSTRUCTION OF KEY VARIABLES

J.B.1. CALL DURATION, RESOLUTION, AND CUSTOMER SATISFACTION

Our firm and associated subcontractors track call resolution, average call duration, and customer satisfaction at the agent-month level. Customer satisfaction is collected by randomly sending surveys to customers who have interacted with contact center agents. From the survey results, our firm calculates a monthly average agent customer satisfaction score. Not all customers complete these surveys, so customer satisfaction is calculated monthly rather than at the chat level. Call resolution is also calculated at the agent-month level. To calculate call resolution, our data firm also uses an algorithm that incorporates elements of chat text and future interactions between the customer

and data firms to calculate a monthly average call resolution score. Although our firm calculates an agent-month average monthly call duration when tracking agent performance, our chat-level dataset allows us to calculate call duration data for each chat in our sample. However, because call resolution is only available on a monthly basis, we report our omnibus productivity measure at the agent-month level.

J.B.2. MEASURING AGENT SKILL, FIRM, AND TENURE

We measure the tenure of the agent in months of work experience. The turnover of the contact center is extremely high. Our firm considers agents with over 6 months of experience to be “very experienced” while agents during their first two months are considered to be “in training.” In our regression specifications, we control for agent tenure using a set of fixed effects for each month of agent tenure. We do not count weeks when agents are not actively working due to vacation or management practices like staggered scheduling intended to reduce worker burnout.

Our regressions also include controls for each agent’s firm, which is the company or subcontractor employing each agent. Agents are employed directly by our data firm as well as by four other business process outsourcing firms. For instance, the firm name for an outsourced worker is “Convergys,” one of our outsourcing firms, while the company of a directly employed call center agent is “data firm.”

We also include controls for agent location—the physical location where the agent is employed. For example, several of our subcontractors have locations in Cebu City, so many agents are located in Cebu. While many of the US-based call center agents are employed directly by our data firm, the outsourcing firms also employ US-based workers. Contact center agents based in the US tend to be clustered in cities with a low cost of living, such as Hazard, Kentucky, and Reno, Nevada. Outside of the United States, most workers are based in the Philippines. As a result of high turnover, contact center firms frequently open and close locations.

We construct an agent skill index that incorporates call handling speed, issue resolution rates, and customer satisfaction. For each firm, we rank agents based on three monthly performance indicators: average handle time (lower times receive higher ranks), call resolution rate, and customer satisfaction (higher rates receive higher ranks). The agent’s rank is calculated within firm. Data firm agents are ranked against their peers, while the Convergys agents are compared with other Convergys employees. These rankings are then averaged into a single skill index at the agent-month level. We then categorize agents into quintiles based on their average skill index from the previous quarter. The skill index incorporates call speed and quality, while comparing individuals within the same organization.

J.B.3. PRODUCTIVITY

Our omnibus productivity measure is monthly resolutions per hour. To calculate this, we divide the share of calls resolved (the monthly call resolution score) by the number of calls handled per hour. Although call duration is available for each chat, resolution data is only available at the agent-month level. Consequently, we report our productivity results at the agent-month level.

J.B.4. CUSTOMER SENTIMENT

The text-based nature of customer support sheds light on the job experience of contact center agents. Although a call may ultimately be resolved or receive a high customer satisfaction score, customers often start conversations feeling stressed, frustrated, or angry. Regularly dealing with complaints and dissatisfied customers is emotionally taxing and contributes to high worker turnover. Conversations also tend to be fairly long, close to 40 minutes, and often involve complex topics such as managing payroll for employees, issues connecting with banks, and calculating taxes.

To capture the experience of customer-agent interactions, we use natural language processing to analyze the language, context, and emotional content of chat transcripts. We employ SiEBERT, an LLM fine-tuned for sentiment analysis on numerous datasets, including product reviews and tweets, with similar syntax and content to that of chat-based customer support. This model has demonstrated robustness to noisy, real-world data, such as social media posts or customer reviews. SiEBERT uses surrounding text to capture word meaning, which has been shown to outperform other methods of sentiment analysis (Hartmann et al., 2023).

For each piece of text, the model produces a sentiment score measured on a scale from -1 to 1 , where -1 indicates negative sentiment and 1 indicates positive. We separately calculate sentiment scores for agents’ text (agent sentiment) and customers’ text (customer sentiment). We then aggregate these chat-level variables into measures of average agent sentiment and average customer sentiment for each agent-month.

J.B.5. LANGUAGE COMPREHENSIBILITY AND FLUENCY

We use Gemini Pro, a large language model (LLM), to measure comprehensibility and native fluency (Gemini Team, 2024). Our criteria for native-like fluency are based on the Interagency Language Roundtable (ILR) “functionally native” language proficiency standard, adapted for written text. The ILR is an organization comprising federal government agencies that coordinate and share information on foreign language activities. Functionally Native Proficiency, the highest level on the ILR scale, describes language ability at which an individual communicates with complete fluency

and precision on all levels normally pertinent to professional needs; displays cultural understanding equivalent to a native speaker; and demonstrates the ability to counsel, persuade, and negotiate in the language as effectively as a well-educated native speaker.

Using the prompt below, we ask the LLM to score each agent’s text from a scale of 1 to 5, where 1 = Definitely not a native American English speaker; 2 = Probably not a native American English speaker; 3 = Uncertain if native American English speaker; 4 = Probably a native American English speaker; and 5 = Definitely a native American English speaker. The full prompt is:

You are given a conversation transcript from a customer service agent helping a customer that only includes what the customer service agent writes to the customer. Based on the transcript of agents’ conversations, provide a score that determines if the customer service agent is likely a native speaker of American English. For the native speaker assessment, look for traits such as correct grammar based on standard American English conventions; use of American English vocabulary, idioms, and phrasing rather than that of other dialects; natural, fluent-sounding language that does not appear stilted or translated; adherence to American cultural norms and reference points in word choice and examples; whether the writer can produce written material with the proficiency of a highly educated native speaker, demonstrating a superior command of the language. For each excerpt, provide a score from 1 to 5 for the native speaker assessment, where 1 = Definitely not a native American English speaker; 2 = Probably not a native American English speaker; 3 = Uncertain if native American English speaker; 4 = Probably a native American English speaker; and 5 = Definitely a native American English speaker. Only return your score of native speaker assessment. Do not include an explanation.

To validate the LLM’s scores, we had two independent human reviewers evaluate 100 randomly selected agent conversations. The reviewers did not have access to each other’s scores or the LLM results. The mean score given by the LLM was 4.29, while the average score from the human evaluators was 4.22. The difference between these scores was not statistically significant, with a p-value of 0.22.

The comprehensibility score captures the general fluency and ease of understanding in the responses of the customer service agents, ranging from very difficult to comprehend (1) to very fluent and easily understandable (5). The score assesses the clarity of communication, frequency and impact of errors, and general language proficiency demonstrated in the writing, regardless of whether the writer is a native speaker. A Philippines-based agent may have a very high level of fluency, but not speak like a native speaker. For instance, a common sign-off is to tell customers to “have a

blessed day.” While that phrase is not a colloquial English phrase, it is grammatically correct and fully comprehensible. The full prompt is:

You are given a conversation transcript from a customer service agent helping a customer that only includes what the customer service agent writes to the customer. Based on the transcript of agents’ conversations, score the overall fluency of each excerpt from 1 to 5, regardless of whether it seems to be written by a native speaker, where 1 = Very difficult to comprehend, multiple errors impeding comprehension; 2 = Somewhat difficult to understand, some errors impeding comprehension; 3 = Mostly understandable but with some errors; 4 = Fluent and understandable with only minor errors that do not impact meaning; and 5 = Very fluent and easily understandable with no significant errors. Only return your score of fluency. Do not include an explanation.

We conducted a similar validation exercise with a random sample of 100 agent conversations, asking human evaluators to assess the comprehensibility of the agents’ speech. The mean comprehensibility score assigned by the LLM was 4.31, while the average score given by human evaluators was 4.40. The difference between these scores was not statistically significant, with a p-value of 0.12.

J.B.6. CONVERSATION TOPIC

Conversations between agents and customers are fairly complex interactions, often discussing details of tax filings, setting up sick leave, firing an employee, or correcting login issues. We use Gemini Pro to capture the topic of each conversation (Gemini Team, 2024).

First, we select a random sample of 5,000 conversations. Using the prompt below, we first ask the LLM to extract the topic of the conversation in one to three words, following a procedure similar to Choi et al. (2024). We then use the LLM to cluster these extracted topics into 50 distinct topic groups, each with a concise single-sentence definition, and validate these categories with contact center personnel. Then, we classify each of the chat conversations into a topic category, “other” or “unsure” using an LLM. Appendix Figure A.XIV displays a breakdown of the most common chat topics encountered by agents. Conversation topics in our data are highly skewed, with the two most common topics—payroll and taxes and account access and management issues—making up 50 percent of all chat topics. The next five topics account for the next 25 percent of conversations. In total, the top 16 topics account for over 90 percent of chats in the data. This pattern is consistent across customer support; generally a small number of fairly common issues make up the bulk of customer inquiries. The full prompt is:

The following is the text of a chat between a customer service agent and a customer who has reached out to the Intuit customer support team. The customer is usually a small business owner based in the US. Based on the transcript, categorize the topic of the conversation into a one-to-three-word topic. If the topic is about a customer not being able to log into their account, categorize the chat as “Login.” If the conversation is about paying employees, the correct topic is “Payroll.” If the conversation is about paying suppliers, the correct topic is “Supplier Payments”. If the topic is about tax-related issues, the correct topic is “Taxes.” If the topic is about ensuring that two sets of records (usually the balances of two accounts) are in agreement, the right topic is “Reconciliation.” When analyzing the transcript, do not focus on details like the identity of the customer, filler greeting text, metadata around the agent joining the conversation, or variables that obscure sensitive details like \$SSN, \$EMAIL, \$NAME-1, \$ADDRESS, hyperlink metadata like
, or case numbers. The output format should be: topic, problem type, issue resolution, probability resolved, skill required, covid-19 related.

Once we have a list of 50 categories and a clear definition, we then ask the LLM to classify the topic of each conversation into one of the 50 categories. The LLM could also classify a chat as “other” if the conversation does not fit any of the designated categories, or “unsure” if the LLM is unable to classify the conversation into any specific category.

All together, we are able to classify 85 percent of all conversations into a topic category. The subjects follow a highly skewed distribution—almost 50 percent of contact center questions fall into payroll & taxes or account access & management.

We validate the LLM-generated topic categorizations by employing three independent human evaluators to classify a random sample of 100 conversations into our predefined topic categories. In 30 percent of the sample, the three human evaluators unanimously agree on the same topic (“unanimous topic”). In 75 percent of the sample (which includes the unanimous subset), at least two evaluators agree on a topic (“modal topic”). The remaining 25 percent of the sample consists of discordant conversations in which each evaluator selects a different topic. When comparing the LLM’s topic selections to these human-evaluated subsets, we find that for unanimous-consensus conversations, the LLM selects the same topic as the human evaluators 87 percent of the time. For modal-consensus conversations, the LLM selects the modal topic 66 percent of the time. In the case of discordant conversations, where there is no agreement between human reviewers, the LLM’s selected topic matches one of the three distinct human-selected topics 74 percent of the time.

We use these topic categories to rank topics by overall topic frequency across all conversations. We also categorize topics according to their overall frequency with respect to an individual agent.

J.B.7. CONVERSATION SENTIMENT

Sentiment analysis involves determining the emotions or attitudes expressed in a piece of text. The valence of unstructured text data is widely used in consumer marketing, predicting stock market returns, measuring consumer sentiment, monitoring social media, and understanding voting behavior.

Machine learning-based methods, particularly those utilizing transfer learning models, generally outperform other sentiment classification approaches. [Hartmann et al. \(2023\)](#) performs a meta-analysis of more than 1,100 experimental results, demonstrating that transfer-learning models are superior for sentiment classification. They also provide an open-sourced, fine-tuned language model, SiEBERT, incorporating transfer-learning best practices, which we use for our sentiment analysis. The model generates a sentiment measure on a scale from -1 to 1, with -1 indicating negative sentiment and 1 indicating positive sentiment.

In our analysis, we classify sentiment separately for agents’ speech and customers’ text in each conversation. Customer sentiment reflects the emotional experience of the 40-minute or so chat-based interaction, while agent sentiment captures the tone of the agents’ responses.

J.B.8. CONVERSATION SIMILARITY

We create textual embeddings of agent-customer conversations and compare similarity of these embeddings across workers and over time. Textual embeddings take a given body of text and transform it into a high-dimensional vector that represents its “coordinates” in linguistic space. Two pieces of text will have more similar coordinates if they share a common meaning or style. The specific embedding given to a body of text will depend on the embedding model used. We form our text embeddings using all-MiniLM-L6-v2, an LLM that is specifically intended to capture and cluster semantic information to assess similarity across text ([Hugging Face, 2023](#)). Once we create an embedding for each conversation, we can compare the similarity of conversations by looking at the cosine similarities of their associated vectors; this common approach yields a score of 0 if two pieces of text are semantically orthogonal and a score of 1 if they have the same meaning ([Koroteev, 2021](#)). For context, the sentences “Can you help me with logging in?” and “Why is my login not working?” have a cosine similarity of 0.68 in our model.

J.C. EMPIRICAL SPECIFICATIONS

J.C.1. PRE-TREATMENT WORKER SKILL SPECIFICATION

$$y_{it} = \delta_t + \alpha_i + \sum_{q=1}^5 \beta_q (AI_{it} \times Q_{iq}) + \gamma X_{it} + \epsilon_{it} \quad (1)$$

Our worker skill specification allows us to estimate how the impact of assistance varies by agent skill when they receive access to AI assistance. The regression is conducted at the agent-month level, where:

- y_{it} is the resolutions per hour of agent i in year-month t .
- δ_t represents year-month fixed effects.
- α_i denotes agent-level fixed effects.
- AI_{it} is an indicator equal to 1 if agent i has access to AI assistance at time t , 0 otherwise.
- Q_{iq} is an indicator function that equals 1 if agent i belonged to skill quintile q at the time of treatment, where q ranges from 1 (lowest skill) to 5 (highest skill).
- X_{it} is a set of time-varying controls, specifically fixed effects for agent tenure in months.
- β_q represents the average treatment effect of AI assistance for agents in skill quintile q

We estimate this equation separately for each of our outcome variables, which include our main measure of productivity for agent i in year-month t (resolutions per hour), as well as call resolution rate, customer satisfaction score, average call duration, calls handled per hour, and requests to speak to the manager. We cluster standard errors at the agent level to account for within-agent correlations in the error terms. In our main analysis, we find very similar results across estimators and similar main effects across adoption cohorts, so we estimate this regression specification using OLS.

J.C.2. PRE-TREATMENT WORKER TENURE SPECIFICATION

$$y_{it} = \delta_t + \alpha_i + \sum_{e=1}^5 \beta_e (AI_{it} \times Exp_{ie}) + \epsilon_{it} \quad (2)$$

Our tenure specification allows us to estimate how the impact of assistance varies by agent experience when they receive access to AI assistance. The regression is conducted at the agent-month level, where:

- y_{it} is the resolutions per hour of agent i in year-month t .
- δ_t represents year-month fixed effects.
- α_i denotes agent-level fixed effects.
- AI_{it} is an indicator equal to 1 if agent i has access to AI assistance at time t , 0 otherwise.
- Exp_{ie} is an indicator that equals 1 if agent i has e months of experience at the time of treatment, and 0 otherwise. We divide months of experience into five categories: agents in their first month on the job (“0 months”), 1-2 months of experience, 3-6 months of experience, 7-12 months, and more than 12 months of experience.
- X_i includes a set of fixed effects for agent i ’s skill quintile at the time of treatment. This is only available for treated agents and is time-invariant.
- β_e is the average treatment effect of AI assistance for agents with e months of experience when treated.

We estimate this equation separately for each of our outcome variables, which include our main measure of productivity for agent i in year-month t (resolutions per hour), as well as call resolution rate, customer satisfaction score, average call duration, and calls handled per hour. The skill quintile at AI treatment is time invariant and is not defined for control group workers. Standard errors are clustered at the agent level, and we estimate this regression using OLS.

J.C.3. ADHERENCE TO AI RECOMMENDATIONS

$$y_{it} = \delta_t + \alpha_i + \sum_{a=1}^5 \beta_a (AI_{it} \times Adh_{ia}) + \gamma X_{it} + \epsilon_{it} \quad (3)$$

This specification allows us to estimate how the impact of AI assistance varies by treated agents’ adherence in their first month of AI access. The regression is conducted at the agent-month level, where:

- y_{it} is the resolutions per hour of agent i in year-month t .
- δ_t represents year-month fixed effects.
- α_i denotes agent-level fixed effects.
- AI_{it} is an indicator equal to 1 if agent i has access to AI assistance at time t , 0 otherwise.

- Adh_{ia} is an indicator equal to 1 if agent i is in the a th quintile of adherence in their first month of access, 0 otherwise.
- X_{it} includes time-varying controls, specifically fixed effects for agent tenure.
- β_a is the average impact of AI assistance for agents in the a th quintile of initial adherence.

Standard errors are clustered at the agent level. We estimate this regression with OLS.

J.C.4. HETEROGENEITY BY CHAT TOPIC

$$y_{itc} = \delta_t + \alpha_i + \sum_{r=1}^4 \beta_r (AI_{it} \times Topic_{cr}) + \gamma X_{itc} + \epsilon_{itc} \quad (4)$$

This topic-based specification allows us to estimate how the impact of AI assistance varies by the routine nature of customers’ problems. The regression is conducted at the chat level, where:

- y_{itc} is the duration of chat c assigned to agent i in year-month t .
- δ_t represents year-month fixed effects.
- α_i denotes agent-level fixed effects.
- AI_{it} is an indicator equal to 1 if agent i has access to AI assistance at time t , 0 otherwise.
- $Topic_{cr}$ is an indicator equal to 1 if chat c belongs to topic frequency category r (where r ranges from 1 to 4), 0 otherwise. The frequency of the topic is defined using frequency across all chats and agents.
- X_{it} includes fixed effects for agent tenure and the overall category of topic frequency ($Topic_{cr}$).
- β_r estimates the impact of AI assistance on chat duration for chats in the topic frequency category r .

Standard errors are clustered at the agent level. We estimate this regression with OLS. We drop the small number of topics that are classified as unsure or “other.” We also estimate a separate specification on the agent-specific frequency of the technical support issue.

$$y_{itc} = \delta_t + \alpha_i + \sum_{r=1}^4 \beta_r (AI_{it} \times AgentTopic_{icr}) + \gamma X_{cit} + \epsilon_{itc} \quad (5)$$

- $AgentTopic_{icr}$ is an indicator equal to 1 if chat c belongs to agent-specific topic frequency category r (where r ranges from 1 to 4), 0 otherwise. Topic frequency is defined relative to conversations conducted by agent i .

- X_{cit} includes time-varying controls, specifically fixed effects for agent tenure, aggregate topic category defined over all conversations ($Topic_{cr}$), and agent-specific topic frequency category ($AgentTopic_{icr}$).
- β_r estimates the impact of AI assistance on chat duration for chats in agent-specific topic frequency category r .

J.C.5. ATTRITION

$$attrit_{it} = \delta_t + \beta AI_{it} + \gamma X_{it} + \epsilon_{it} \quad (6)$$

This specification allows us to look at how AI assistance impacts agent attrition. The regression is conducted at the agent-month level, where:

- $attrit_{it}$ is equal to 1 if agent i leaves in year-month t .
- δ_t represents year-month fixed effects.
- AI_{it} is an indicator equal to 1 if agent i has access to AI assistance at time t , 0 otherwise.
- X_{it} includes time-varying controls, specifically fixed effects for agent tenure, agent location, country and company employing the agent (data firm or subcontractor).
- β_a is the average impact of AI assistance on attrition.

Attrition captures both voluntary or involuntary separations, which are undistinguishable in our data. Standard errors are clustered at the agent level, and we estimate this regression with OLS. We cannot include for agent-fixed effects, because agents only leave once, so agent-fixed effects are collinear with attrition. We drop all observations for treated agents before treatment because, by construction, agents must survive through treatment to receive AI assistance. We also estimate specifications where we estimate the attrition effects by agent tenure and skill at AI deployment.

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